

MARIAN COLLEGE KUTTIKANAM AUTONOMOUS

UG Programme (Honors) Degree October 2025 Regular Examinations (I Semester UG)

First Semester

251CCRUBC102 Discrete Mathematics

Time: 2 Hours

Maximum Marks: 40

Answer one question from each bunch

BUNCH I

I.A ✓ For each of these collections of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises.

a) "If I wake up early, I will exercise."

"I woke up early."

b) "If it's a holiday, I will relax."

"I did not relax."

c) "If you eat healthy, you will feel energetic."

"If you feel energetic, you will work out."

d) "Either she will come to the party, or she will stay home."

"She is not staying home."

(CO1:8 Marks)

OR

I.B (a) Consider the following proposition:

"If a student studies hard, then they will pass the exam."

Is this true if the student studies hard but does not pass? Is this false if the student does not study but passes?

(b) Find the contrapositive, converse and inverse of the following proposition.

"If a person is a teenager, then they are between 13 and 19 years old"

(c) Evaluate the truth value of the statement:

"The sky is blue and $2 + 2 = 5$."

(CO1:8 Marks)

BUNCH II

II.A

1. Is the function $f(x) = x + 1$ from the set of integers to the set of integers a bijection? Justify.

2. Let g be the function from $\{a, b, c, d\}$ to $\{1, 2, 3, 4\}$ with $g(a) = 4$, $g(b) = 2$, $g(c) = 1$, and $g(d) = 3$. Is g a bijection? Justify.

(CO2:8 Marks)

OR

II.B ✓ Let $A = \{x \mid x \text{ is an integer such that } 1 \leq x \leq 10\}$ and $B = \{x \mid x \text{ is an integer such that } 5 \leq x \leq 15\}$. Find (a) $A \cup B$ (b) $A \cap B$ (c) $(A \cup B)'$ (d) $A - (B \cup (A \cap B))$. (CO2:8 Marks)

BUNCH III

III.A Determine whether the following relations are reflexive, symmetric, anti symmetric and transitive.

1. aRb if and only if a and b have same birth month
2. aRb if and only if vehicles a and b have same registration district code

(CO3:8 Marks)

OR

III.B Examine the following relations on $\{1, 2, 3, 4\}$ and determine which of them are reflexive, symmetric, antisymmetric and transitive:

1. $R_1 = \{(1, 1), (1, 2), (2, 1)\}$
2. $R_2 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$

(CO3:8 Marks)

BUNCH IV

IV.A Consider a small logistics company that is responsible for transporting packages across three cities: City X, City Y and City Z. The company has three types of trucks, each capable of carrying different amounts of cargo and requiring different amounts of fuel for each route. The table below represents the amount of cargo each truck can carry to each city, as well as the amount of fuel required for each route.

The matrix A below shows the cargo capacity and fuel consumption for each type of truck:

$$A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}. \text{ Find the inverse of the matrix.}$$

(CO4:8 Marks)

OR

IV.B Consider a small logistics company that is responsible for transporting packages across three cities: City X, City Y and City Z. The company has three types of trucks, each capable of carrying different amounts of cargo and requiring different amounts of fuel for each route. The table below represents the amount of cargo each truck can carry to each city, as well as the amount of fuel required for each route.

The matrix A below shows the cargo capacity and fuel consumption for each type of truck:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ -2 & -4 & -5 \end{bmatrix}. \text{ Find the inverse of the matrix.}$$

(CO4:8 Marks)

BUNCH V

V.A ✓ Test the consistency of the following system of linear non homogenous equations. If so, find the solution or solutions,

$$3x + 5y + 6z = 7$$

$$x + 3y - 2z = 5$$

$$2x + 4y + 3z = 8$$

(CO5:8 Marks)

OR

V.B.1 A commodity was produced by using 3 units of labour and 2 units of capital, the total cost is Rs.62. If the commodity had been produced by using 4 units of labour and 1 unit of capital, the cost is Rs.56. Construct and solve the linear equations of the problem. (CO5:4 Marks)

V.B.2 Solve the system of linear equations.

$$x + 3y = 8$$

$$2x - y = 9$$

(CO5:4 Marks)

CO1	Apply rules of inferences for propositional logic to solve logical and mathematical problems
CO2	Demonstrate comprehension of fundamental concepts related to sets, set operations, and functions
CO3	Analyze the relations to classify them based on their properties
CO4	Analyze matrices, their operations, determinants, inverses, and ranks
CO5	Solve the system of linear equations